

Title: Development of an IoT-Based Smart Energy Monitoring and Predictive Maintenance System for University Laboratories

ABSTRACT

University laboratories use computers, air-conditioning units, lighting systems, laboratory equipment, and other electrical devices that consume significant amounts of energy. In many institutions, electricity usage is monitored only through monthly utility bills, making it difficult to identify equipment with excessive consumption or possible electrical faults. This study proposes the development of an Internet of Things-based smart energy monitoring and predictive maintenance system for university laboratories. The system will use electrical sensors, microcontrollers, and a web-based dashboard to collect and display real-time data on voltage, current, power consumption, temperature, and equipment operating duration. It will also analyze historical readings to identify unusual energy usage and possible equipment maintenance needs. The proposed system aims to help laboratory administrators reduce energy waste, detect abnormal electrical behavior, and improve equipment maintenance planning.

Keywords: Internet of Things, smart energy monitoring, predictive maintenance, electrical engineering, sensors, microcontrollers, energy efficiency, anomaly detection, university laboratories, real-time monitoring

CHAPTER 1 INTRODUCTION

Background of the Study

Energy consumption is an important concern for universities because classrooms, offices, laboratories, and other facilities use a large number of electrical devices. University laboratories may contain desktop computers, servers, air-conditioning units, printers, projectors, testing machines, power supplies, and specialized equipment. These devices may operate for several hours every day and contribute significantly to the institution's electricity expenses.

Traditional energy monitoring usually depends on monthly utility bills or manual inspection of electrical meters. These methods provide general information about total electricity consumption but may not show which laboratory, device, or equipment causes unusually high energy usage. As a result, administrators may have difficulty identifying energy waste, faulty equipment, or electrical devices that are operating longer than necessary.

The Internet of Things allows sensors and electronic devices to collect and exchange information through a network. In an energy monitoring system, electrical sensors can measure voltage, current, power, energy consumption, and equipment temperature. A microcontroller can process the collected data and send it to a database or online platform. A dashboard can then display the information in real time.

Predictive maintenance uses data to identify signs that equipment may require inspection, repair, or replacement. For example, unusual increases in power consumption, temperature, or operating time may indicate that a machine is no longer functioning efficiently. Detecting these patterns early may help prevent equipment failure and reduce maintenance costs.

This study proposes an IoT-based smart energy monitoring and predictive maintenance system for university laboratories. The system will collect electrical and environmental data, display real-time readings, record historical usage, and detect abnormal patterns that may indicate energy waste or possible equipment problems.

Statement of the Problem

University laboratories use different types of electrical equipment, but many institutions do not have a centralized system for monitoring the energy consumption and operating condition of individual devices.

Without real-time monitoring, administrators may not immediately notice unusual electricity usage, overheating equipment, devices left operating after laboratory hours, or changes that may indicate possible equipment failure.

This study seeks to answer the following questions:

1. What electrical and operational data should be collected from laboratory equipment?
2. How can IoT sensors and microcontrollers be used to monitor energy consumption in real time?
3. How can the system identify unusual energy usage and possible maintenance concerns?
4. How accurate and reliable is the proposed monitoring system?
5. How usable is the web-based dashboard for laboratory administrators and maintenance personnel?

Objectives of the Study

General Objective

The study aims to develop an IoT-based smart energy monitoring and predictive maintenance system for university laboratories.

Specific Objectives

The study aims to:

1. Design a system that collects voltage, current, power, energy consumption, temperature, and operating-time data.
2. Develop an IoT device using sensors and a microcontroller.
3. Create a database for storing real-time and historical energy readings.

4. Develop a web-based dashboard for viewing laboratory and equipment data.
5. Detect abnormal energy consumption and possible equipment maintenance concerns.
6. Evaluate the system in terms of accuracy, reliability, functionality, and usability.
7. Recommend energy-saving and maintenance actions based on the collected information.

Significance of the Study

University Administrators

The system may help administrators understand how energy is consumed across selected laboratories. The information may support decisions related to electricity budgeting, equipment replacement, and sustainability programs.

Laboratory Personnel

Laboratory personnel may use the system to check whether equipment is operating normally. They may also receive alerts when unusual consumption or temperature readings are detected.

Maintenance Personnel

Maintenance teams may use historical data and alerts to identify equipment that requires inspection. This may help improve preventive and predictive maintenance planning.

Students and Faculty

Students and faculty may benefit from safer and more reliable laboratory equipment. The project may also increase awareness of responsible energy use.

Future Researchers

The study may serve as a reference for future projects involving IoT, smart buildings, energy management, electrical monitoring, equipment diagnostics, and predictive maintenance.

Scope and Delimitation

The proposed system will focus on selected electrical equipment in university laboratories. It will monitor:

- Voltage
- Current
- Power consumption
- Total energy consumption
- Equipment temperature
- Operating duration
- Equipment status

The system will use sensors connected to a microcontroller. The collected data will be transmitted to a database and displayed through a web-based dashboard.

The proposed system may generate alerts for:

- Unusually high power consumption
- Equipment operating outside scheduled hours
- High temperature readings
- Sudden changes in electrical behavior
- Possible maintenance requirements

The study will not replace professional electrical inspection or maintenance diagnosis. Its alerts will only serve as early warnings based on collected data. The prototype will be tested using selected laboratory devices and may not cover the entire university electrical system.

Definition of Terms

Anomaly Detection

The process of identifying readings or behavior that differ significantly from normal equipment operation.

Current

The flow of electrical charge through a circuit, commonly measured in amperes.

Internet of Things

A network of sensors, devices, and systems that collect and exchange data through wired or wireless communication.

Microcontroller

A programmable electronic device used to receive sensor readings, process data, and control system functions.

Predictive Maintenance

A maintenance approach that uses equipment data and operating patterns to identify possible problems before failure occurs.

Power Consumption

The rate at which electrical energy is used by a device, commonly measured in watts.

Real-Time Monitoring

The continuous collection and display of data with minimal delay.

Smart Energy Monitoring

The use of connected sensors and software to measure, record, and analyze electricity consumption.

CHAPTER 2

REVIEW OF RELATED LITERATURE

Internet of Things in Energy Management

The Internet of Things is widely used in smart homes, industrial systems, agriculture, transportation, and building management. IoT systems connect sensors, microcontrollers, communication networks, databases, and user applications.

In energy management, IoT devices can collect data from electrical equipment and transmit it to a monitoring platform. This allows users to observe consumption patterns without manually checking each device. IoT-based monitoring may also support remote access, automated alerts, and historical analysis.

Smart Energy Monitoring Systems

Smart energy monitoring systems provide detailed information about electricity usage. Unlike traditional utility meters, these systems may monitor individual equipment, rooms, or electrical circuits.

The collected information can help users identify:

- Devices with high energy consumption
- Peak usage periods
- Equipment operating outside scheduled hours
- Sudden changes in energy behavior
- Opportunities for reducing electricity waste

Real-time energy monitoring may improve decision-making because administrators do not have to wait for the monthly utility bill before identifying possible problems.

Electrical Sensors and Microcontrollers

Electrical monitoring systems commonly use sensors to measure voltage, current, power, and energy consumption. Temperature sensors may also be used to monitor the operating condition of electrical devices.

Microcontrollers such as Arduino and ESP32 can receive and process sensor readings. Microcontrollers with Wi-Fi capability can transmit the collected information to a server or cloud-based database.

The accuracy of the system depends on proper sensor calibration, stable network communication, correct wiring, and reliable data processing.

Predictive Maintenance

Predictive maintenance focuses on monitoring equipment condition to determine when maintenance may be needed. Unlike corrective maintenance, which happens after equipment failure, predictive maintenance attempts to identify warning signs before a breakdown occurs.

Possible warning signs may include:

- Increased electricity consumption
- Unusual temperature readings
- Irregular operating patterns
- Longer operating duration
- Repeated abnormal readings
- Sudden changes in electrical load

Predictive maintenance may help reduce downtime, avoid unexpected failures, and improve equipment life.

Anomaly Detection

Anomaly detection is used to identify data that differs from normal patterns. In an energy monitoring system, an anomaly may be a sudden increase in current, unusually high temperature, or energy consumption outside the expected range.

A simple system may use threshold-based detection. For example, the system may generate an alert if the temperature or power consumption exceeds a predefined limit.

A more advanced system may use historical averages, standard deviation, machine learning, or time-series analysis to identify unusual behavior.

Web-Based Monitoring Dashboards

A web-based dashboard allows users to view sensor data through a browser. The dashboard may show:

- Current equipment status
- Real-time electrical readings
- Daily and weekly consumption
- Historical charts
- Equipment alerts
- Estimated energy costs
- Maintenance recommendations

A clear dashboard is important because users must be able to understand the information quickly. Charts, labels, filters, and warning indicators may improve system usability.

Energy Efficiency in University Facilities

Universities consume electricity through academic buildings, laboratories, offices, libraries, and common facilities. Energy efficiency programs may include switching off unused equipment, improving operating schedules, replacing inefficient devices, and monitoring consumption.

A smart monitoring system may support these efforts by providing evidence about when, where, and how electricity is being used.

Conceptual Framework

The proposed system follows an input-process-output framework.

Inputs:

- Voltage readings
- Current readings
- Power consumption
- Energy consumption
- Temperature
- Operating duration
- Equipment schedule
- Equipment information

Processes:

- Sensor data collection
- Data transmission
- Data validation
- Database storage
- Historical comparison
- Anomaly detection
- Alert generation
- Dashboard visualization

Outputs:

- Real-time energy readings
- Historical energy reports
- Abnormal usage alerts
- Equipment status
- Maintenance warnings
- Energy-saving recommendations

CHAPTER 3

METHODOLOGY

Research Design

The study will use a design and development research approach. This approach is appropriate because the study involves designing, building, testing, and evaluating a working prototype.

The development process will include:

1. Requirements analysis
2. System design
3. Hardware development
4. Software development
5. Sensor calibration
6. System integration
7. Prototype testing
8. User evaluation

System Architecture

The proposed system will contain the following components:

Sensor Layer

The sensor layer will collect electrical and environmental readings from selected equipment. It may include:

- Current sensor
- Voltage sensor
- Energy monitoring sensor
- Temperature sensor

Processing Layer

A microcontroller will receive the sensor readings, perform initial processing, and prepare the data for transmission.

Communication Layer

The microcontroller will transmit data through Wi-Fi or another available network connection.

Database Layer

The database will store:

- Faculty or administrator accounts

- Laboratory information
- Equipment records
- Sensor readings
- Historical consumption
- Alert records
- Maintenance notes

Application Layer

The web-based application will display real-time readings, charts, alerts, reports, and maintenance information.

Proposed Hardware Components

The proposed prototype may use:

- ESP32 microcontroller
- Current sensor
- Voltage sensor
- Temperature sensor
- Energy meter module
- Relay module, if remote control is included
- Power supply
- Electrical enclosure
- Laboratory test equipment

Proposed Software Components

The system may use:

- Microcontroller programming environment
- Web application framework
- Database management system
- Application programming interface
- Charting or data visualization library
- Local or cloud server

Data Collection

The system will collect readings at regular intervals. Each record may contain:

- Equipment ID
- Laboratory name
- Voltage
- Current
- Power

- Total energy consumption
- Temperature
- Date and time
- Equipment status

Historical readings will be used to determine the normal operating range of each monitored device.

Anomaly Detection Method

The initial prototype will use threshold-based and historical comparison methods.

The system may generate an alert when:

- Power consumption exceeds the configured limit
- Temperature exceeds the safe operating limit
- Equipment operates beyond its expected schedule
- Current usage changes significantly from the normal range
- Repeated abnormal readings are detected

For future improvement, the system may use machine learning or predictive analytics to identify more complex equipment behavior.

System Development Procedure

Phase 1: Requirements Analysis

The researcher will identify the electrical equipment to be monitored and consult laboratory personnel regarding common energy and maintenance concerns.

Phase 2: System Design

The researcher will prepare the hardware architecture, database design, system flow, dashboard layout, and alert rules.

Phase 3: Prototype Development

The sensors and microcontroller will be assembled and programmed. The database and web dashboard will also be developed.

Phase 4: Integration

The hardware and software components will be connected. Sensor readings will be transmitted to the database and displayed on the dashboard.

Phase 5: Testing

The system will be tested using selected laboratory equipment. The researcher will compare sensor readings with a reference measuring device where possible.

Phase 6: Evaluation

Selected laboratory personnel, maintenance staff, and administrators will evaluate the system.

System Evaluation

The proposed system will be evaluated based on:

- Functional suitability
- Accuracy
- Reliability
- Usability
- Performance efficiency
- Security
- Maintainability

Testing Procedures

Functional Testing

This will determine whether the sensors, database, dashboard, reports, and alerts operate according to the requirements.

Accuracy Testing

Sensor readings will be compared with readings from a reference meter or measuring instrument.

Connectivity Testing

The system will be tested under stable and interrupted network conditions.

Alert Testing

The researcher will simulate abnormal readings to determine whether the system produces the correct warnings.

Usability Testing

Selected users will perform common tasks such as viewing equipment status, checking historical consumption, reviewing alerts, and reading maintenance recommendations.

Data Analysis

The following methods may be used:

- Frequency
- Percentage
- Mean
- Error percentage
- Response time
- System uptime
- User evaluation scores

For accuracy testing, the recorded sensor readings may be compared with reference readings.

Ethical and Safety Considerations

Electrical safety procedures will be followed during hardware installation and testing. The prototype will be installed or supervised by individuals with appropriate electrical knowledge.

The system will not automatically disconnect critical laboratory equipment unless proper authorization and safety mechanisms are available.

User account information and system records will be protected through authentication and access control.

EXPECTED RESULTS

The proposed system is expected to:

1. Display real-time energy consumption for selected laboratory equipment.
2. Store historical energy usage data.
3. Identify unusually high electrical consumption.
4. Generate temperature and energy alerts.
5. Show equipment operating duration.
6. Support maintenance planning through early warnings.
7. Help administrators identify energy-saving opportunities.
8. Provide a user-friendly web dashboard.

PROPOSED SYSTEM FEATURES

- Faculty and administrator login
- Laboratory and equipment registration
- Real-time electrical monitoring
- Equipment status display
- Daily and weekly energy reports
- Historical consumption charts
- Temperature monitoring
- Abnormal usage alerts

- Predictive maintenance warnings
- Equipment maintenance records
- Estimated energy cost
- Exportable reports

PROPOSED PROJECT SCHEDULE

Month 1

- Requirements gathering
- Literature review
- Identification of laboratory equipment

Month 2

- Hardware and system design
- Database design
- Dashboard planning

Month 3

- Sensor and microcontroller development
- Web application development

Month 4

- Hardware and software integration
- Sensor calibration

Month 5

- Testing and evaluation
- Data analysis

Month 6

- System improvement
- Documentation
- Final presentation

CONCLUSION

The proposed IoT-based smart energy monitoring and predictive maintenance system aims to improve energy management in university laboratories. By combining electrical sensors, microcontrollers, database technology, anomaly detection, and a web-based dashboard, the system may provide administrators with better information about equipment consumption and operating condition.

The project is expected to support energy efficiency, early fault detection, maintenance planning, and responsible use of university resources.

REFERENCES

References will be added based on the final technical literature, electrical engineering standards, IoT studies, energy-monitoring research, and predictive-maintenance sources used in the actual study.